

Enhancing Question Answering Explainability with QED and LLaMA-based Fine-tuning

1 Introduction

Recent advances in large language models (LLMs) have demonstrated impressive performance on a wide range of natural language understanding and generation tasks. However, when it comes to question answering (QA), most systems still operate as black boxes, providing answers without transparent explanations of their underlying reasoning. The lack of explainability poses significant challenges for both users and researchers, particularly in applications where trust, reliability, and debuggability are essential.

The QED (Question Explanation Dataset) framework (Lamm et al., 2021) offers a structured way to make QA more explainable by breaking down the process into clear steps: identifying a supporting sentence, extracting the answer span, and linking referential equalities between question and passage. While this method improves explanation quality and user trust, prior work has mostly used specialized models and training, so it remains unknown whether instruction-based LLMs can reach similar results with prompting alone.

Addressing this gap, our research investigates whether modern instruction-tuned LLMs can effectively generate QED-style explanations, and how different prompting strategies influence their performance. This question is particularly timely, as instruction-based LLMs are increasingly being used as general-purpose tools for complex tasks, but there has been limited systematic evaluation of their performance on structured explanation tasks such as QED.

Prior work on QED has reported strong results for models such as SpanBERT and T5 on explanation generation, but these models rely on architectural modifications and extensive supervised fine-tuning with expert-annotated data. This makes it challenging to adapt or extend such approaches in settings where resources or domain-specific data are limited.

Question: who wrote the film howl's moving castle?
Passage: Howl's Moving Castle is a 2004 Japanese animated fantasy film written and directed by Hayao Miyazaki. It is based on the novel of the same name, which was written by Diana Wynne Jones. The film was produced by Toshio Suzuki.
Answer: Hayao Miyazaki
(1) Sentence Selection Howl's Moving Castle is a 2004 Japanese animated fantasy film written and directed by Hayao Miyazaki.
(2) Referential Equality the film howl's moving castle = Howl's Moving Castle
(3) Entailment X is a 2004 Japanese animated fantasy film written and directed by ANSWER. $\vdash$ ANSWER wrote X.

Figure 1: QED explanations decompose the question-passage relationship in terms of referential equality and predicate entailment. (Lamm et al., 2021)

In contrast, instruction-based LLMs offer a flexible and potentially scalable alternative, requiring only natural language prompts to adapt to new tasks.

Our approach leverages instruction-following models and prompt-based learning to perform the joint answer and explanation prediction task defined by the QED framework. We systematically compare multiple prompting strategies, evaluate model performance both before and after lightweight parameter-efficient fine-tuning, and benchmark our results against previously reported baselines. Through this comparative analysis, we aim to provide new insights into the strengths and limitations of prompt-based structured explanation generation, inform best practices for leveraging LLMs in explainable QA, and identify avenues for future research.

In this study, we address the following research questions:

RQ1: Can modern prompt-based LLMs achieve comparable QED task performance to the QED paper's structured models?

RQ2: How do different few-shot prompting strategies influence QED performance for base and fine-tuned models?

2 Related Work

Recent work in explainable QA has emphasized explanations that are truly grounded in evidence, not just answers alone. The QED framework (Lamm et al., 2021), built on Natural Questions (Kwiatkowski et al., 2019), exemplifies this shift by breaking explanations into interpretable, semantically anchored steps that connect the question and passage. In doing so, QED bridges the gap between free-form rationales and fully formal logic-based explanations. Earlier work explored structured models such as SpanBERT (Joshi et al., 2020), equipped with coreference resolution capabilities (Pradhan et al., 2012), and large-scale sequence-to-sequence models like T5 (Raffel et al., 2020) for generating QED explanations. These models demonstrated that it is possible to learn the QED framework directly from data, producing explanations that closely reflect the underlying passage evidence. However, these approaches rely heavily on supervised training, architectural specialization, and task-specific fine-tuning.

The rise of prompt-based large language models has made it possible to generate explanations with minimal retraining, using methods like chain-of-thought prompting to encourage more transparent reasoning. However, even as these models make explanation generation more accessible, their outputs do not always align with the evidence in the text, and can sometimes result in convincing yet unsupported rationales (Wei et al., 2022). This gap between fluency and faithfulness has motivated a broader search for techniques that can ensure explanations are not just plausible, but truly grounded in the facts.

To further improve faithfulness, recent research has explored verification-based and structured explanation strategies that build on the same grounding principles as QED. For example, chain-of-verification (Dhuliawala et al., 2023) requires each reasoning step to be checked against the actual evidence, while retrieval-guided counterfactual generation (Paranjape et al., 2021) tests explanations by systematically altering key facts. Alongside these trends, frameworks like ExplaGraphs (Saha et al., 2021) and e-SNLI have expanded the notion of explanation beyond QED’s single-sentence focus,

exploring graph-structured and free-text justifications. Together, these lines of work highlight the ongoing balance between interpretability, robustness, and the need to ground explanations tightly in the decision process.

Our approach improves upon prior methods by adapting LLaMA-based and Qwen models with targeted prompts and minimal post-processing to generate QED-style explanations, achieving faithfulness and interpretability without retraining or complex pipelines. By directly evaluating with QED’s metrics, we show that this approach makes explanation generation easier, and lets us directly compare how clear and trustworthy prompt-based models are, compared to more complex supervised systems. This can help bring reliable explanations to more users.

3 Methods

3.1 Dataset

For all experiments in this study, we use the QED dataset (Lamm et al., 2021), which is based on a subset of the Google Natural Questions corpus and annotated by experts to enable systematic evaluation of question answering and explanation. Each example consists of a question, a Wikipedia passage, an answer span, and a QED explanation that identifies the supporting sentence, extracts the answer, and links relevant noun phrases between the question and passage, ensuring each explanation is tightly grounded in the source text.

QED provides high-quality linguistic coverage, with 77% of examples containing full QED explanations, 23% with only valid answers, and about 10% lacking a valid short answer. We restrict our experiments to the 77% of cases with complete QED explanations, ensuring that all evaluated examples include the full annotation needed for structured explanation generation.

A distinctive feature of QED is its careful annotation of different types of referential expressions, including proper names (such as "How I Met Your Mother"), definite noun phrases (for example, "the president of the United States"), generics (such as "a dead zone"), and pronouns (like it, they, he, she). The dataset also provides explicit treatment of bridging references, where the relevant entity is implied rather than directly stated in the text. This detailed coverage supports structured explanation generation and analysis of model reasoning.

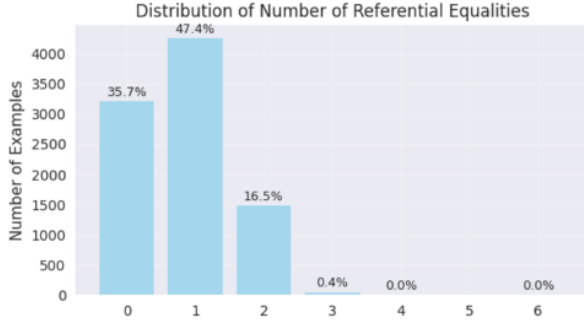


Figure 2: Distribution of the number of referential equalities per example in the QED dataset.

3.2 Task Definition

Given a question $q = (q_1, \dots, q_m)$, a passage title $t = (t_1, \dots, t_k)$, and a passage $c = (c_1, \dots, c_n)$, the model predicts a QED annotation $\hat{y} = \langle \hat{s}, \hat{a}, \hat{e} \rangle$, where:

- $\hat{s}$: the supporting sentence from the passage.
- $\hat{a}$: the answer span(s) in that sentence, possibly coreference-resolved.
- $\hat{e}$: pairs of phrases showing which parts of the question and sentence refer to the same thing.

The task is defined as learning a function

$$f : (q, t, c) \longrightarrow (\hat{a}, \hat{s}, \hat{e}),$$

We train the model to minimize the difference between its predictions and the gold annotations across all examples:

$$\text{Optimize} : \sum_i \ell(\text{gold}_i, \text{prediction}_i)$$

where ℓ measures how well the model’s prediction matches the correct answer and explanation for each example.

3.3 Models

To address our research questions, we selected two instruction-based language models: LLaMA-3-8B-Instruct (AI@Meta, 2024) and Qwen2.5-7B-Instruct (Yang et al., 2024). These models were chosen for their comparable size, open-source availability, and strong performance on reasoning tasks. Their similar parameter counts allow us to directly compare architectural differences under equivalent computational constraints. Both are instruction-tuned, making them well-suited for structured tasks

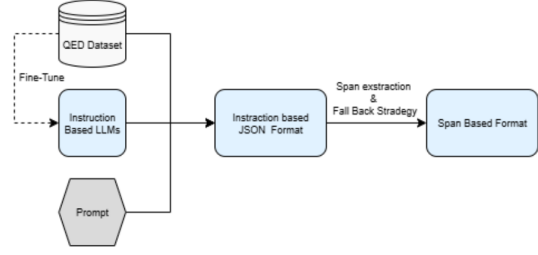


Figure 3: QED data is used for prompting and LoRA fine-tuning of instruction-tuned LLMs. Models generate explanations in a JSON schema, which are parsed, validated, and mapped to text spans for QED evaluation.

that require following detailed prompts, such as the QED framework.

LLaMA-3-8B-Instruct is Meta’s instruction-tuned model, built on the LLaMA-3 architecture. It is specifically optimized for producing structured outputs and handling multi-step reasoning, which aligns well with the requirements of the QED explanation task. Qwen2.5-7B-Instruct, developed by Alibaba, represents the latest version of the Qwen series and is also designed for robust instruction-following and explanation generation. Its distinct architecture provides an opportunity to examine how different model designs approach the generation of structured explanations.

Each model was evaluated in two configurations. The first is the base setting, where models are used in their original, instruction-tuned state without any further adaptation to the QED data. The second involves parameter-efficient fine-tuning using the LoRA method, which adds a small number of trainable parameters to the base model while keeping the main weights fixed. This approach is particularly effective under computational resource constraints. We applied 4-bit quantization to further reduce memory requirements, and performed fine-tuning over three epochs using QED training examples. To encourage diverse exposure to QED annotation patterns and minimize bias from fixed demonstration prompts, the fine-tuning process included two randomly selected examples in each prompt.

3.4 Prompting Framework

To evaluate how demonstration strategies influence model performance, we systematically vary the use and content of few-shot examples within each prompt. Each prompt begins with a fixed instruction section that clearly describes the task, speci-

fies the required JSON output schema, and defines formatting expectations. The main experimental variable is the type and number of demonstrations included before the actual test instance.

We consider six prompting strategies. The first is the zero-shot approach, where the model receives only task instructions and no demonstration examples. The next two strategies include a single fixed demonstration: “first only,” which uses a carefully selected example that highlights bridging and implicit referential phenomena, and “second only,” which features explicit entity resolution and pronoun use. The “both” strategy combines these two fixed examples to ensure the prompt covers a broader range of linguistic challenges.

In addition to fixed demonstrations, we include two randomization-based strategies. “Random one” selects a single example at random from the demonstration pool, while “random two” draws two random examples. These settings help us test whether specific example selection influences model performance and whether models can generalize from diverse demonstration patterns.

Our initial experiments showed that relying solely on bridging examples could lead models to overproduce bridge annotations. To address this, we used prompt balancing-combining examples that represent both bridging and non-bridging referential phenomena in the “both” strategy. This balanced approach exposes models to multiple annotation types within a single prompt, helping us understand whether they can learn effectively from varied demonstration styles.

All prompts maintain a clear and consistent structure, allowing for fair comparison across prompting strategies.

3.5 Evaluation Metrics

We assess model performance using the same metrics as the original QED study, enabling direct comparison with previous work. **Answer accuracy** is computed as the percentage of questions for which the predicted answer span achieves at least a 70% character-level F1 overlap with the gold-standard span. This threshold accommodates minor boundary mismatches while maintaining strict span-level precision. The overlap ratio is computed using character-level F1, defined as:

$$F1 = \frac{tp}{tp + \frac{fp+fn}{2}}$$

where tp is the number of overlapping charac-

ters between predicted and gold spans, fp is the number of predicted characters not in the gold span, and fn is the number of gold characters not in the prediction.

Referential equality is evaluated along two dimensions: (1) *Mention Identification*, which measures the model’s ability to detect referential expressions in both questions and passages, and (2) *Mention Alignment*, which assesses whether the model correctly pairs referentially equivalent expressions between questions and their corresponding evidence sentences, including complex cases such as bridging references.

3.6 Output Processing and Error Handling

Given the variability in outputs produced by large language models, we implement a robust processing pipeline to ensure reliable evaluation. Our system first attempts to extract JSON objects directly from the model’s response, prioritizing well-formatted code blocks but also searching for JSON fragments using regular expressions if needed. When the output does not meet the required format due to issues such as missing braces or extra punctuation, fallback strategies attempt to recover partial information so that as much data as possible can still be evaluated. Cases that remain ambiguous or unreadable are logged for manual review and error analysis.

After extracting the structured output, we align the predicted answer, sentence, and referential equalities with the original passage text. This process computes exact character offsets for each span, handles multi-word and nested entities, and verifies that all boundaries correspond to valid text in the source. Since LLM outputs may differ from the ground truth in phrasing or contain minor errors, our span matching process uses case-insensitive and punctuation-normalized comparison, along with fuzzy string matching, to maximize recall without sacrificing accuracy.

This multi-stage approach ensures that our evaluation remains comprehensive, even when models deviate from the strict schema or make small mistakes, and that our results reflect the underlying performance of each system as accurately as possible.

4 Experimental Evaluation

All evaluations were conducted using the QED dataset development set. The evaluation frame-

Model	Mention Identification			Mention Alignment			Ans. Acc. (%)
	P	R	F1	P	R	F1	
SpanBERT	77.6	64.4	70.4	68.9	57.2	62.5	74.5
T5	70.3	72.3	71.3	58.3	59.9	59.1	79.2
LLaMa-3	12.7	12.4	12.6	5.6	5.4	5.5	70.3
Qwen2.5	20.8	21.2	21.0	11.8	12.0	12.0	74.9
LLaMa-3 FT	44.1	36.1	39.7	26.0	21.2	23.4	54.8
Qwen2.5 FT	40.4	21.4	28.0	25.3	13.4	17.5	68.2

Table 1: Performance comparison across QED evaluation metrics for baseline structured models versus LLMs in base and fine-tuned conditions. All models uses to random prompt strategy, including the fine tune. All models were evaluated using the "two random" prompt strategy, which was also used during fine-tuning.

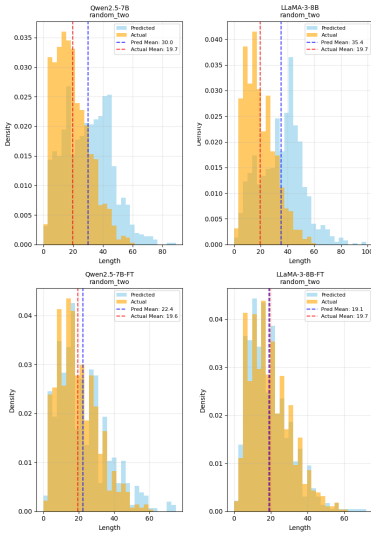


Figure 4: Histograms of question-reference lengths (predicted vs. gold). The top row shows base models, while the bottom row shows fine-tuned models, which better calibrate toward the gold distribution.

work employed multiple overlap thresholds with non-strict matching modes, with the chosen threshold of 0.7 for all models rather than the original 0.9 on the QED paper to account for LLMs’ tendency to extend outputs beyond exact answer boundaries. Each prompting strategy was tested across both model architectures in base and fine-tuned conditions. JSON output parsing utilized the automated pipeline described in the methodology, with problematic predictions logged for error analysis. All experiments used official QED evaluation metrics to maintain comparability with original SpanBERT and T5 baseline results

4.1 LLM vs Structured Models Performance (RQ1)

The comparison between baseline structured models and LLMs reveals a substantial performance

gap in the core components of the QED framework. While base LLMs exhibit competitive answer accuracy (Qwen2.5: 74.9%, LLaMA-3: 70.3%), their performance on structured explanation tasks—specifically mention identification and mention alignment—is markedly lower, with F1 scores significantly trailing those of structured baselines.

Fine-tuning leads to considerable improvements: LLaMA-3 reaches 39.7% F1 in mention identification (a $3.15\times$ increase) and 23.4% in mention alignment (a $4.25\times$ increase), while Qwen2.5 demonstrates more modest but consistent gains. However, these improvements frequently come at the expense of answer accuracy, which declines post-tuning. This trade-off between explanation quality and answer extraction highlights a critical area for further investigation in current fine-tuning approaches.

Notably, these gains were achieved using only three epochs of training under constrained computational resources, indicating that even modest increases in training resources may lead to further performance improvements. Nonetheless, both fine-tuned models remain well below the structured baselines on the QED metrics, highlighting the limitations of current instruction-tuning techniques. Prompt-based inference alone proves inadequate for the structured demands of the QED framework. While LLMs may produce plausible answers, they often fail to align with the precise semantic and referential constraints required by the QED formalism. These findings point to the need for more advanced optimization strategies or architecture-level modifications to enable LLMs to generate high-fidelity, rule-conformant explanations.

To further understand the impact of fine-tuning, we analyze question reference length distributions (Figure 4). Base models over-extend spans, whereas fine-tuning shifts predictions toward the gold distribution. This calibration increases pre-

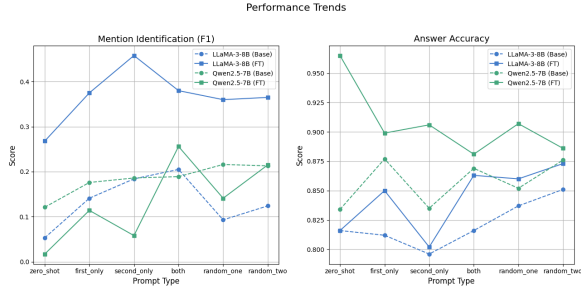


Figure 5: Performance trends across prompting strategies for mention identification (left) and answer accuracy (right), comparing base and fine-tuned models.

cision in Mention Identification by reducing over-inclusion, improves boundary localization leads to a raising F1 and yields cleaner, non-overlapping anchors. These gains achieved after only 3 LoRA epochs under limited compute, suggest that additional epochs could deliver further improvements. Nevertheless, performance still trails structured baselines, indicating that prompt-only methods are insufficient and that more targeted training remains necessary.

4.2 Prompting Strategy Effects (RQ2)

We examine how prompting strategies, varying in both quantity (few-shot approaches) and quality (static versus random examples) affect performance across base and fine-tuned models (Figure 5).

Quantity Effects. Few-shot prompting consistently improves Mention Identification performance relative to zero-shot baselines. However, increasing the number of examples does not uniformly enhance performance: LLaMA-FT achieves peak performance with a single predefined example (second_only condition) and shows degraded performance when provided with multiple examples. This suggests interference effects rather than cumulative learning benefits for this model.

Quality Effects. The type of demonstration example appears less influential than the number. When comparing single-shot strategies (first_only, second_only, and random_one conditions), no approach demonstrated consistent superiority across models, with random examples occasionally outperforming static ones. This suggests that generalization in example usage may enhance model performance. The both strategy shows strong Mention Identification performance by incorporating diverse referential equality phenomena, including both bridging and non-bridging examples, enabling models to learn complex referential behav-

iors. However, no prompting approach showed consistent superiority in answer accuracy across all models.

For Mention Identification, fine-tuned LLaMA outperformed all other models across prompting conditions, indicating that fine-tuning enhances referential equality detection capabilities. In terms of answer accuracy, fine-tuned versions consistently outperformed their base counterparts, with Qwen-FT achieving significantly superior performance across all prompt types. Notably, this model exhibited an inverse relationship between the number of few-shot examples and accuracy: zero-shot prompting yielded exceptionally high performance (0.965), which decreased as additional examples were provided. This counterintuitive pattern suggests that fine-tuning optimized the model so specifically for this task that additional contextual information introduced interference rather than guidance. Overall, no single model or prompting strategy demonstrated consistent superiority across all evaluation metrics, indicating that QED tasks in LLMs present greater complexity than initially anticipated.

5 Conclusion

This study investigated whether instruction-based language models can generate high-quality, structured QED explanations. Our experiments show that while models like LLaMA-3 and Qwen2.5 are able to achieve answer accuracy close to traditional baselines, they still fall short in producing precise and reliable explanations, particularly when it comes to referential alignment and mention identification. Fine-tuning these models led to meaningful improvements in explanation quality, but did not fully bridge the gap with structured approaches such as SpanBERT and T5.

These results suggest that, although instruction-tuned language models offer flexibility and require less task-specific adaptation, there are still clear challenges in using them for structured explanation tasks. Continued progress will likely depend on combining prompt-based learning with methods that can enforce stronger semantic and referential consistency. Our work provides a benchmark for future research in explainable question answering and underscores the importance of developing models that not only answer correctly but also explain their reasoning in a clear and grounded way.

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